```java
package Homework;
import java.util.Scanner;

public class Lab1Ex1 {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter n: ");
        int n = sc.nextInt();

        if (n < 1 || n > 1000) {
            System.out.println(x: "Invalid input. Please enter a number from 1 to 1000.");
            sc.close();
            return;
        }

        int sum = 0;
        System.out.print(s: "Output: ");

        for (int k = 1000; k >= n; k--) {
            System.out.print(k);
            if (k > n) {
                System.out.print(s: ", ");
            }
            sum += k;
        }
        double avg = (double) sum / (1000 - n + 1);

        System.out.println();
        System.out.println("The sum of 1000 to " + n + " is: " + sum);
        System.out.println("Avg is: " + avg);
        sc.close();
    }

}
```



```
pimea@Macs-MacBook-Pro-3 Java %  /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-27.jdk/Contents/Home/bin/java -XX:+ShowCodeDeta
ilsInExceptionMessages -cp /Users/pimea/Library/Application\ Support/Code/User/workspaceStorage/09dff92b973638dc1fb511a109c57160/re
dhat.java/jdt_ws/Java_a1fab877/bin Homework.Lab1Ex1
Enter n: 560
Output: 1000, 999, 998, 997, 996, 995, 994, 993, 992, 991, 990, 989, 988, 987, 986, 985, 984, 983, 982, 981, 980, 979, 978, 977, 976, 975, 974, 973, 972, 971, 970, 969, 968, 967, 966, 965, 964, 963, 96
2, 961, 960, 959, 958, 957, 956, 955, 954, 953, 952, 951, 950, 949, 948, 947, 946, 945, 944, 943, 942, 941, 940, 939, 938, 937, 936, 935, 934, 933, 932, 931, 930, 929, 928, 927, 926, 925, 924, 923, 922
, 921, 920, 919, 918, 917, 916, 915, 914, 913, 912, 911, 910, 909, 908, 907, 906, 905, 904, 903, 902, 901, 900, 899, 898, 897, 896, 895, 894, 893, 892, 891, 890, 889, 888, 887, 886, 885, 884, 883, 882,
 881, 880, 879, 878, 877, 876, 875, 874, 873, 872, 871, 870, 869, 868, 867, 866, 865, 864, 863, 862, 861, 860, 859, 858, 857, 856, 855, 854, 853, 852, 851, 850, 849, 848, 847, 846, 845, 844, 843, 842,
841, 840, 839, 838, 837, 836, 835, 834, 833, 832, 831, 830, 829, 828, 827, 826, 825, 824, 823, 822, 821, 820, 819, 818, 817, 816, 815, 814, 813, 812, 811, 810, 809, 808, 807, 806, 805, 804, 803, 802, 8
01, 800, 799, 798, 797, 796, 795, 794, 793, 792, 791, 790, 789, 788, 787, 786, 785, 784, 783, 782, 781, 780, 779, 778, 777, 776, 775, 774, 773, 772, 771, 770, 769, 768, 767, 766, 765, 764, 763, 762, 76
1, 760, 759, 758, 757, 756, 755, 754, 753, 752, 751, 750, 749, 748, 747, 746, 745, 744, 743, 742, 741, 740, 739, 738, 737, 736, 735, 734, 733, 732, 731, 730, 729, 728, 727, 726, 725, 724, 723, 722, 721
, 720, 719, 718, 717, 716, 715, 714, 713, 712, 711, 710, 709, 708, 707, 706, 705, 704, 703, 702, 701, 700, 699, 698, 697, 696, 695, 694, 693, 692, 691, 690, 689, 688, 687, 686, 685, 684, 683, 682, 681,
 680, 679, 678, 677, 676, 675, 674, 673, 672, 671, 670, 669, 668, 667, 666, 665, 664, 663, 662, 661, 660, 659, 658, 657, 656, 655, 654, 653, 652, 651, 650, 649, 648, 647, 646, 645, 644, 643, 642, 641,
640, 639, 638, 637, 636, 635, 634, 633, 632, 631, 630, 629, 628, 627, 626, 625, 624, 623, 622, 621, 620, 619, 618, 617, 616, 615, 614, 613, 612, 611, 610, 609, 608, 607, 606, 605, 604, 603, 602, 601, 6
00, 599, 598, 597, 596, 595, 594, 593, 592, 591, 590, 589, 588, 587, 586, 585, 584, 583, 582, 581, 580, 579, 578, 577, 576, 575, 574, 573, 572, 571, 570, 569, 568, 567, 566, 565, 564, 563, 562, 561, 56
0
The sum of 1000 to 560 is: 343980
Avg is: 780.0
pimea@Macs-MacBook-Pro-3 Java %
```

Explorer

Java

Homework

- Lab1Ex1.java
- Lab1Ex2.java
- Lab1Ex3.java
- Lab1Ex4.java
- Lab1Ex5.java
- Lab1Ex6.java
- Task1.class
- Task3.class
- Task5.class

test

Welcome | Lab1Ex3.java | Task1.class | Lab1Ex4.java | Lab1Ex5.java | Lab1Ex6.java | Lab1Ex1.java | Lab1Ex2.java | Task3.class

Homework > Lab1Ex2.java > Lab1Ex2

```java
package Homework;
import java.util.Scanner;

public class Lab1Ex2 {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner (System.in);
        int a,b,c;
        System.out.println (x: "============================");
        System.out.println (x: "Welcome to our program");
        System.out.println (x: "'Quadratic Formula Solver ax^2+bx+c'");
        System.out.println (x: "============================");
        System.out.println (x: "Input a:");
        a = sc.nextInt();
        System.out.println (x: "Input b:");
        b = sc.nextInt();
        System.out.println (x: "Input c:");
        c = sc.nextInt();
        System.out.println (x: "----------------------------------------");

        if (a == 0 && b == 0 && c == 0) {
            System.out.println(x: "The equation has infinite solutions.");
        } else if (a == 0 && b == 0) {
            System.out.println(x: "The equation has no solution.");
        } else if (a == 0) {
            double x = -c / (double)b;
            System.out.println("The root is: " + x);
        } else {}
        double delta = b*b - 4*a*c;

        if (delta > 0) {
            double x1 = (-b + Math.sqrt(delta)) / (2*a);
            double x2 = (-b - Math.sqrt(delta)) / (2*a);
            System.out.println("The roots are: " + x1 + " and " + x2);
        }
        if (delta == 0) {
            double x = -b / (2*a);
            System.out.println("The root is: " + x);
        }
        if (delta < 0) {
            System.out.println(x: "The equation has no real roots.");
        }
    sc.close();


}
}

```

Problems | Output | Debug Console | Terminal | Ports

Run: Lab1Ex2

```
pimea@Macs-MacBook-Pro-3 Java %  /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-27.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/pimea/Library/Application\ Support/Code/User/workspaceStorage/09dff92b973638dc1fb511a109c57160/redhat.java/jdt_ws/Java_a1fab877/bin Homework.Lab1Ex2
============================
Welcome to our program
'Quadratic Formula Solver ax^2+bx+c'
============================
Input a:
2
Input b:
6
Input c:
4
----------------------------------------
The roots are: -1.0 and -2.0
pimea@Macs-MacBook-Pro-3 Java %
```

Explorer

Java

Homework

- Lab1Ex1.java
- Lab1Ex2.java
- Lab1Ex3.java
- Lab1Ex4.java
- Lab1Ex5.java
- Lab1Ex6.java
- Task1.class
- Task3.class
- Task5.class

test

Welcome | Lab1Ex3.java | Task1.class | Lab1Ex4.java | Lab1Ex5.java | Lab1Ex6.java | Lab1Ex1.java | Lab1Ex2.java | Task3.class

Homework > Lab1Ex3.java > Lab1Ex3

```java
package Homework;
import java.util.Scanner;

public class Lab1Ex3 {
    Run | Debug
    public static void main(String[] args) {
        Scanner in = new Scanner (System.in);
        System.out.println(x: "Input 5 numbers:");

        int[] inputtedNumber = new int[5];

        for (int num = 0; num < 5; num += 1) {
            inputtedNumber[num] = in.nextInt();
        }

        System.out.println(x: "Input:");
        for (int num = 0; num < 5; num += 1) {
            System.out.println(inputtedNumber[num]);
        }

        int maxNum = Integer.MIN_VALUE;
        int minNum = Integer.MAX_VALUE;

        for (int checkNum = 0; checkNum < 5; checkNum += 1) {
            maxNum = Math.max(maxNum, inputtedNumber[checkNum]);
            minNum = Math.min(minNum, inputtedNumber[checkNum]);
        }

        System.out.println(x: "Output:");
        System.out.println("The maximum number: " + maxNum);
        System.out.println("The minimum number: " + minNum);
        in.close();
    }
}

```

Problems | Output | Debug Console | Terminal | Ports

Run: Lab1Ex3

```
pimea@Macs-MacBook-Pro-3 Java %  /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-27.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/pimea/Library/Application\ Support/Code/User/workspaceStorage/09dff92b973638dc1fb511a109c57160/redhat.java/jdt_ws/Java_a1fab877/bin Homework.Lab1Ex3
Input 5 numbers:
12 32 560 -20 2
Input:
12
32
560
-20
2
Output:
The maximum number: 560
The minimum number: -20
pimea@Macs-MacBook-Pro-3 Java %
```

Explorer

Java

Homework

- Lab1Ex1.java
- Lab1Ex2.java
- Lab1Ex3.java
- Lab1Ex4.java
- Lab1Ex5.java
- Lab1Ex6.java
- Task1.class
- Task3.class
- Task5.class

test

Welcome | Lab1Ex3.java | Task1.class | Lab1Ex4.java | Lab1Ex5.java | Lab1Ex6.java | Lab1Ex1.java | Lab1Ex2.java | Task3.class

Homework > Lab1Ex4.java > Lab1Ex4

```java
package Homework;
import java.util.Scanner;


public class Lab1Ex4 {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner (System.in);
        int yyyy, mm, dd;
        System.out.println(x: "Input your birth date:");
        System.out.println(x: "Input Year:");
        yyyy = sc.nextInt();
        System.out.println(x: "Input Month:");
        mm = sc.nextInt();
        System.out.println(x: "Input Day:");
        dd = sc.nextInt();

        System.out.println(yyyy + "-" + mm + "-" +dd + " (Year/Month/Day)");
        if (yyyy < 1900 & yyyy > 2026) {
        System.out.println(x: "Output: Invalid");
        } else if (mm < 1 & mm > 12) {
        System.out.println(x: "Output: Invalid");
        }else if (dd < 1 & dd > 31) {
        System.out.println(x: "Output: Invalid");
        } else {
        System.out.println(x: "Output: Valid");
        sc.close();
        }

    }
}
```

Problems | Output | Debug Console | Terminal | Ports

Run: Lab1Ex4

```
pimea@Macs-MacBook-Pro-3 Java %  /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-27.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/pimea/Library/Application\ Support/Code/User/workspaceStorage/09dff92b973638dc1fb511a109c57160/redhat.java/jdt_ws/Java_a1fab877/bin Homework.Lab1Ex4
Input your birth date:
Input Year:
2002
Input Month:
6
Input Day:
23
2002-6-23 (Year/Month/Day)
Output: Valid
pimea@Macs-MacBook-Pro-3 Java %
```

Explorer

Java

Homework

- Lab1Ex1.java
- Lab1Ex2.java
- Lab1Ex3.java
- Lab1Ex4.java
- Lab1Ex5.java
- Lab1Ex6.java
- Task1.class
- Task3.class
- Task5.class

test

Welcome | Lab1Ex3.java | Task1.class | Lab1Ex4.java | Lab1Ex5.java | Lab1Ex6.java | Lab1Ex1.java | Lab1Ex2.java | Task3.class

Homework > Lab1Ex5.java > Lab1Ex5

```java
package Homework;
import java.util.Scanner;

public class Lab1Ex5 {
    public static boolean isPrime(int num) {
        if (num <= 1) {
            return false;
        }

        for (int i = 2; i <= Math.sqrt(num); i++) {
            if (num % i == 0) {
                return false;
            }
        }
        return true;
    }

    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Input n: ");
        int testNum = sc.nextInt();

        System.out.println("Prime numbers between 2 and " + testNum + ":");

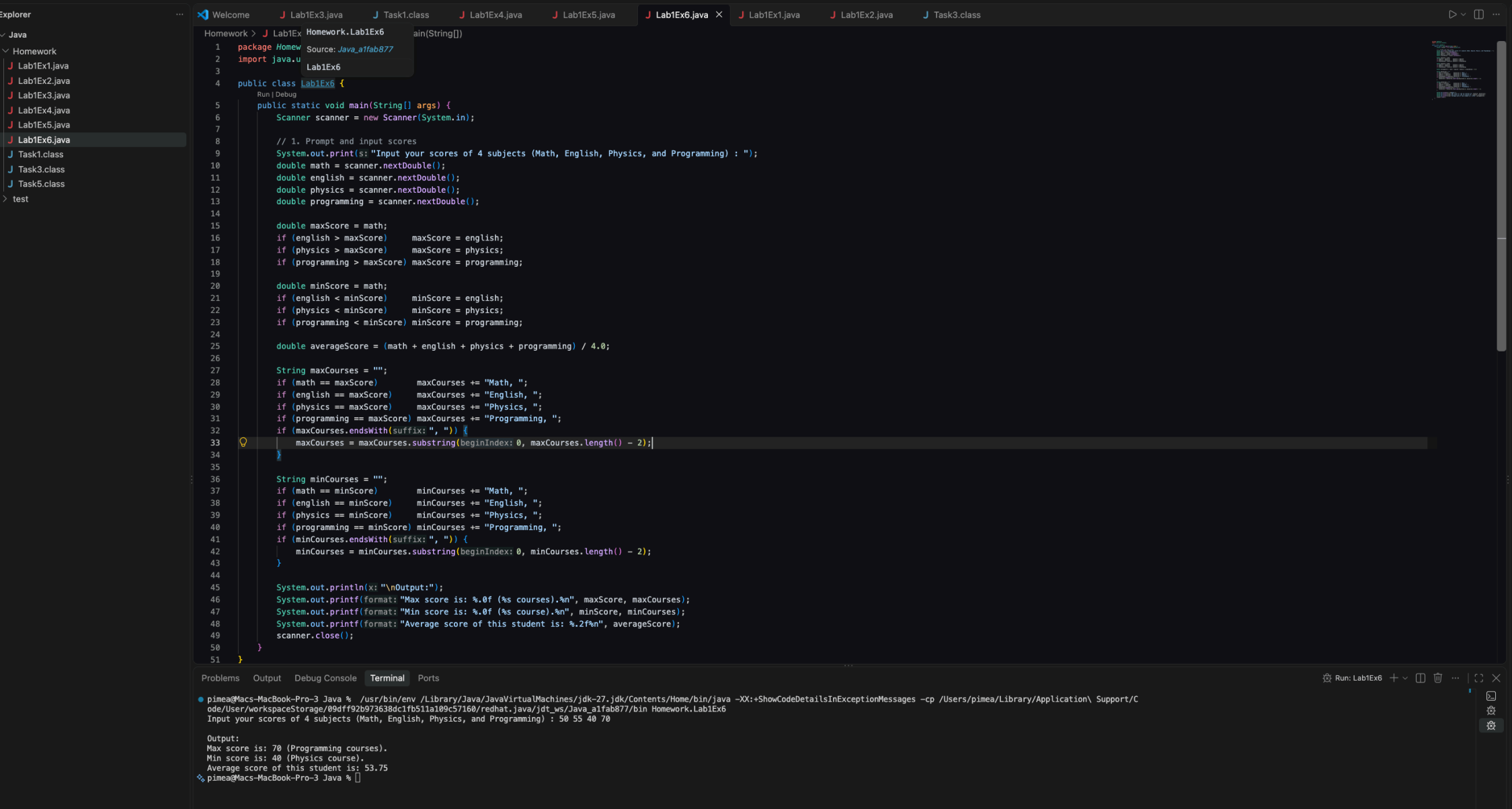
        for (int num = 2; num <= testNum; num++) {
            if (isPrime(num)) {
                System.out.print(num + " ");
            }
        }

        System.out.println();
        sc.close();
    }
}
```

Problems | Output | Debug Console | Terminal | Ports

Run: Lab1Ex5

```
pimea@Macs-MacBook-Pro-3 Java %  /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-27.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/pimea/Library/Application\ Support/Code/User/workspaceStorage/09dff92b973638dc1fb511a109c57160/redhat.java/jdt_ws/Java_a1fab877/bin Homework.Lab1Ex5
Input n: 423
Prime numbers between 2 and 423:
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97 101 103 107 109 113 127 131 137 139 149 151 157 163 167 173 179 181 191 193 197 199 211 223 227 229 233 239 241 251 257 263 269 271 277 281 283 293 307 311 313 317 331 337 347 349 353 359 367 373 379 383 389 397 401 409 419 421
pimea@Macs-MacBook-Pro-3 Java %
```


Explorer
Java
Homework
Lab1Ex1.java
Lab1Ex2.java
Lab1Ex3.java
Lab1Ex4.java
Lab1Ex5.java
Lab1Ex6.java
Task1.class
Task3.class
Task5.class
test
Welcome
Lab1Ex3.java
Task1.class
Lab1Ex4.java
Lab1Ex5.java
Lab1Ex6.java
Lab1Ex1.java
Lab1Ex2.java
Task3.class
Homework.Lab1Ex6
Source: Java_a1fab877
Lab1Ex6
package Homew
import java.u
public class Lab1Ex6 {
Run | Debug
public static void main(String[] args) {
Scanner scanner = new Scanner(System.in);
// 1. Prompt and input scores
System.out.print(s: "Input your scores of 4 subjects (Math, English, Physics, and Programming) : ");
double math = scanner.nextDouble();
double english = scanner.nextDouble();
double physics = scanner.nextDouble();
double programming = scanner.nextDouble();
double maxScore = math;
if (english > maxScore) maxScore = english;
if (physics > maxScore) maxScore = physics;
if (programming > maxScore) maxScore = programming;
double minScore = math;
if (english < minScore) minScore = english;
if (physics < minScore) minScore = physics;
if (programming < minScore) minScore = programming;
double averageScore = (math + english + physics + programming) / 4.0;
String maxCourses = "";
if (math == maxScore) maxCourses += "Math, ";
if (english == maxScore) maxCourses += "English, ";
if (physics == maxScore) maxCourses += "Physics, ";
if (programming == maxScore) maxCourses += "Programming, ";
if (maxCourses.endsWith(suffix: ", ")) {
maxCourses = maxCourses.substring(beginIndex: 0, maxCourses.length() - 2);
}
String minCourses = "";
if (math == minScore) minCourses += "Math, ";
if (english == minScore) minCourses += "English, ";
if (physics == minScore) minCourses += "Physics, ";
if (programming == minScore) minCourses += "Programming, ";
if (minCourses.endsWith(suffix: ", ")) {
minCourses = minCourses.substring(beginIndex: 0, minCourses.length() - 2);
}
System.out.println(x: "\nOutput:");
System.out.printf(format: "Max score is: %.0f (%s courses).%n", maxScore, maxCourses);
System.out.printf(format: "Min score is: %.0f (%s course).%n", minScore, minCourses);
System.out.printf(format: "Average score of this student is: %.2f%n", averageScore);
scanner.close();
}
}
Problems
Output
Debug Console
Terminal
Ports
Run: Lab1Ex6
pimea@Macs-MacBook-Pro-3 Java % /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-27.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/pimea/Library/Application\ Support/Code/User/workspaceStorage/09dff92b973638dc1fb511a109c57160/redhat.java/jdt_ws/Java_a1fab877/bin Homework.Lab1Ex6
Input your scores of 4 subjects (Math, English, Physics, and Programming) : 50 55 40 70
Output:
Max score is: 70 (Programming courses).
Min score is: 40 (Physics course).
Average score of this student is: 53.75
pimea@Macs-MacBook-Pro-3 Java %